

EPD Homepage SEO v1

Project: Educational Publications Department homepage prototype

Version: SEO v1

Date: 2026-09-10

Scope: Whole Next.js application

Objective

This update establishes a professional technical SEO baseline for the localized EPD website. It improves crawlability, indexing signals, search-result presentation, social sharing, structured data, and deployment readiness without adding a third-party SEO dependency.

SEO cannot guarantee a ranking position. Rankings also depend on official content quality, backlinks, accessibility, performance, mobile usability, domain reputation, and search-engine evaluation. This implementation supplies the technical foundation those factors require.

Implemented Changes

`lib/seo.ts`

Added the central SEO configuration and URL helpers.

- Added the canonical site origin, configurable through `NEXT_PUBLIC_SITE_URL`.
- Added a fallback origin of `https://www.epd.gov.lk` for local builds.
- Added localized English, Sinhala, and Tamil titles and descriptions.
- Added locale-specific Open Graph locale identifiers.
- Added helpers for locale URLs and alternate-language URLs.
- Centralized values so metadata, sitemap, robots, manifest, and JSON-LD use the same origin.

Production requirement: set `NEXT_PUBLIC_SITE_URL` to the actual public HTTPS domain. Do not leave the fallback if the site is deployed under another domain.

`app/[locale]/layout.tsx`

Replaced the single generic metadata object with locale-aware `generateMetadata`.

Each localized route now receives:

- A localized title.
- A localized description.
- `metadataBase`.
- Canonical URL.
- `hreflang` alternate links for `en`, `si`, and `ta`.
- Application name, author, creator, publisher, and government category metadata.
- Index/follow directives for normal search crawling.
- Googlebot directives allowing large image previews and full snippets.
- Open Graph website metadata.

- Open Graph image metadata using `/images/hero/top-banner.png`.
- Twitter `summary_large_image` metadata.
- Favicon metadata.

The canonical homepage URLs are:

- `/en`
- `/si`
- `/ta`

`app/[locale]/page.tsx`

Added JSON-LD structured data using `next/script`.

The homepage now describes:

- `GovernmentOrganization` for the Educational Publications Department.
- The organization logo and Sri Lanka service area.
- The parent Ministry of Education organization.
- `WebSite` publisher information.
- `WebPage` URL, language, description, and relationship to the organization/site.

This gives search engines machine-readable context about the publisher, official nature, language, and page relationships.

The JSON-LD uses `dangerouslySetInnerHTML` only for `JSON.stringify` output generated internally from trusted application constants and does not render user-supplied HTML.

`app/sitemap.ts`

Added a framework-generated XML sitemap at `/sitemap.xml`.

It includes all supported homepage and sitemap routes for:

- English.
- Sinhala.
- Tamil.

Each entry includes:

- Absolute canonical URL.
- Last-modified timestamp.
- Change frequency.
- Priority.

The homepage is prioritized at `1.0`; the human-readable sitemap page is prioritized at `0.3`.

Before launch, add any future indexable content routes to this sitemap. Do not add API, preview, search-result, or duplicate-filter URLs.

`app/robots.ts`

Added a generated robots policy at `/robots.txt`.

- Allows normal crawlers to access the localized public routes.
- Disallows `/api/` because API endpoints are not search content.
- Disallows `/_next/` because framework internals do not need indexing.
- Publishes the absolute XML sitemap URL.
- Publishes the canonical host.

`app/manifest.ts`

Added a web app manifest at `/manifest.webmanifest`.

- Defines the official application name and short name.
- Adds a description and localized-site start URL.
- Defines standalone display behavior.
- Defines theme and background colors.
- Reuses the favicon as the available application icon.

`app/[locale]/sitemap/page.tsx`

Added page-specific SEO metadata for the existing localized sitemap pages.

- Gives each sitemap page a distinct title and description.
- Adds a canonical URL for `/en/sitemap`, `/si/sitemap`, and `/ta/sitemap`.
- Adds localized `hreflang` links for the sitemap pages.

`.env.local.example`

Added:

```
NEXT_PUBLIC_SITE_URL=https://www.epd.gov.lk
```

This variable controls every absolute SEO URL generated by the application.

SEO Architecture

```
graph TD
    A[NEXT_PUBLIC_SITE_URL] --> B[lib/seo.ts]
    B --> C[metadata]
    B --> D[sitemap]
    B --> E[robots]
    B --> F[JSON-LD]
    C --> G[localized routes: /en, /si, /ta]
```

The implementation uses Next.js Metadata API and Metadata Route conventions. This keeps SEO output server-rendered and available to crawlers without client-side JavaScript execution.

Validation Completed

The following checks passed after implementation:

- `npx tsc --noEmit`
- `npm run lint`
- `npm run build`

The production build generated these SEO routes:

- `/sitemap.xml`
- `/robots.txt`
- `/manifest.webmanifest`

A production runtime smoke test also confirmed that `/en` contains:

- Canonical metadata.
- Sinhala `hreflang` metadata.
- JSON-LD structured data.
- Open Graph title metadata.

The runtime test returned HTTP 200 for both `/robots.txt` and `/sitemap.xml`, and the sitemap contained the English localized URL.

Pre-Launch Requirements

Must complete

1. Set `NEXT_PUBLIC_SITE_URL` to the final official HTTPS origin.
2. Verify that the organization name, parent ministry, logo, contact details, and descriptions are officially approved.
3. Replace placeholder images and mock content with approved public assets.
4. Verify all canonical URLs after deployment behind the final domain and proxy/CDN.
5. Register the sitemap in Google Search Console and Bing Webmaster Tools.
6. Verify ownership of the official domain in the search-engine webmaster consoles.
7. Add the actual official social profile URLs when they are approved.
8. Ensure the production server does not block crawlers through hosting or CDN rules.

Recommended next phase

- Add page-specific metadata for future news, notice, circular, book, and division detail routes.
- Add `Article`, `NewsArticle`, `Book`, `BreadcrumbList`, and `FAQPage` JSON-LD only where the corresponding visible content exists on the page.
- Generate image metadata with approved 1200x630 social images rather than relying only on the hero image.
- Add a dedicated 180x180 or larger application icon and Apple touch icon.
- Add `og:updated_time` or `dateModified` when content has a reliable publication timestamp.

- Add locale-aware `<html lang>` output at the document level for Sinhala and Tamil routes; the current application sets the locale on the localized layout container while the root document remains `en`.
- Add automated tests for metadata, canonical URLs, sitemap entries, robots directives, and translation parity.
- Monitor Search Console coverage, Core Web Vitals, indexed pages, duplicate canonical reports, and structured-data enhancements.

Known Limitations

Technical SEO improves discoverability but does not create authoritative content by itself. The current prototype still has known ranking limitations:

- Much of the data is static mock content.
- Several sections are not fully translated.
- Some visual tiles and quick links are placeholders.
- There are no detail pages for most books, news, notices, or divisions.
- There is no editorial publishing workflow or content freshness system.
- Counter endpoints are unrelated to search ranking and still require rate limiting before public deployment.
- No analytics or Search Console integration was added; those should be introduced only under the organization's privacy and governance requirements.

File Summary

File	Modification
<code>lib/seo.ts</code>	Centralized site origin, localized SEO copy, and URL helpers.
<code>app/[locale]/layout.tsx</code>	Added locale-aware metadata, canonical URLs, <code>hreflang</code> , robots directives, Open Graph, Twitter, and icon metadata.
<code>app/[locale]/page.tsx</code>	Added GovernmentOrganization, WebSite, and WebPage JSON-LD.
<code>app/[locale]/sitemap/page.tsx</code>	Added page-specific sitemap metadata and alternate-language URLs.
<code>app/sitemap.ts</code>	Added generated XML sitemap route.
<code>app/robots.ts</code>	Added generated robots policy.
<code>app/manifest.ts</code>	Added web app manifest metadata.
<code>.env.local.example</code>	Added <code>NEXT_PUBLIC_SITE_URL</code> configuration.
<code>docs/SEO v1.md</code>	Documented the SEO implementation, validation, requirements, and limitations.